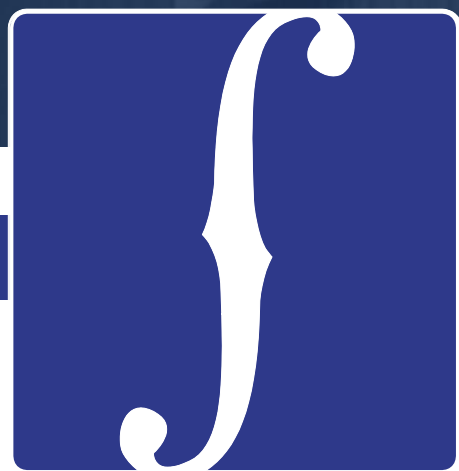




CELLO

CELLO RESEARCH

DECEMBER 2015

- I. The Case for In-House Research
- II. Areas of Focus
- III. The Role of Technology
- IV. Team Bios

THE CASE FOR IN-HOUSE RESEARCH

- **Alpha Generation:** Generate original, consensus-defying ideas through researching homeowner prepayment behavior, MBS pricing models, and industry trends that affect MBS pricing
- **Accuracy:** Street research can oversimplify the complexities of prepayment data. “Paper” trades often do not accurately capture the dynamics of pricing and hedging an MBS trade from inception to close. The Cello Research team serves as a valuable resource in cross-validating/fact-checking Street research.
- **Speed:** Proprietary databases and pricing models help us sift through large datasets very rapidly to analyze the idiosyncratic risk associated with every CMO. Research analysts leverage technology to quickly address queries from the investment team.

AREAS OF FOCUS

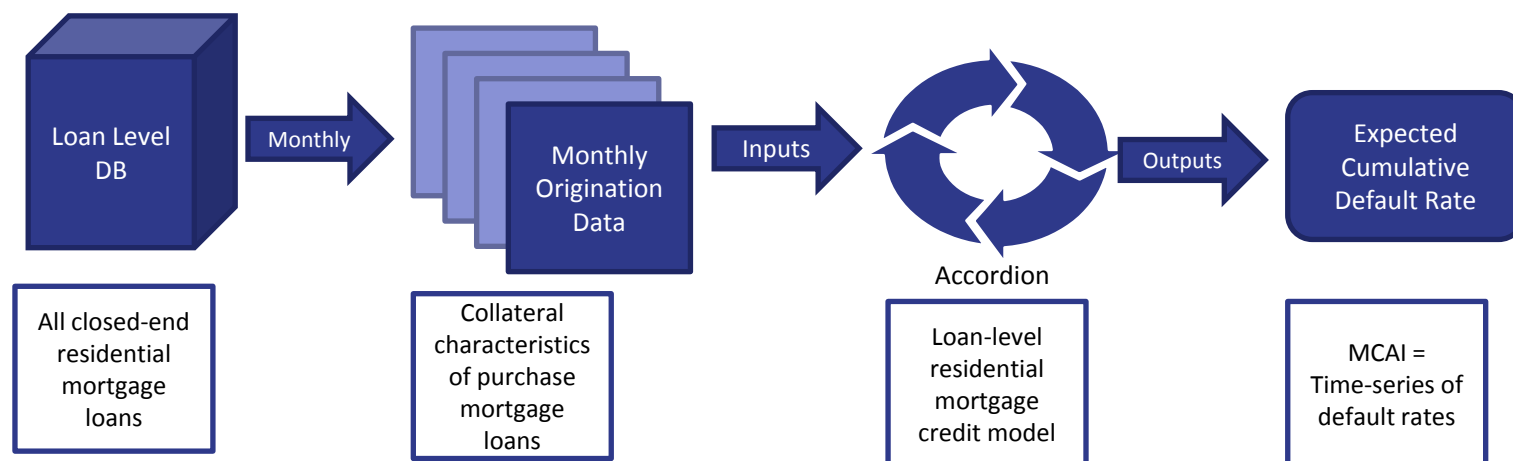
- **Homeowner Prepayment Behavior:** Evolution of Borrower/Loan characteristics, Government housing policy, Mortgage lending industry, Prepayment modeling
- **MBS CMO Valuation and Hedging:** Mortgage pricing models, Interest rate models, Mortgage rate models, House price models
- **Relative Value Models:** Econometric modeling of price/spread relationships for various fixed-income securities
- **Scale (with IT):** Software framework for pricing, hedging and risk management of fixed-income instruments and, in particular, structured products arising in the housing markets

Cello's Mortgage Credit Availability Index (MCAI) assumes that the market increases/decreases credit availability by taking on more/less mortgage default risk.

The fundamental advantage of following this method is that it provides a single comprehensive measure at any point in time that aggregates across the different risk factors that are taken into account in the typical underwriting process (FICO, LTV, DTI, etc.).

The expected default rate is a weighted average of the defaults projected over a "normal" and "stressed" environment.

CONSTRUCTING CELLO'S MORTGAGE CREDIT AVAILABILITY INDEX

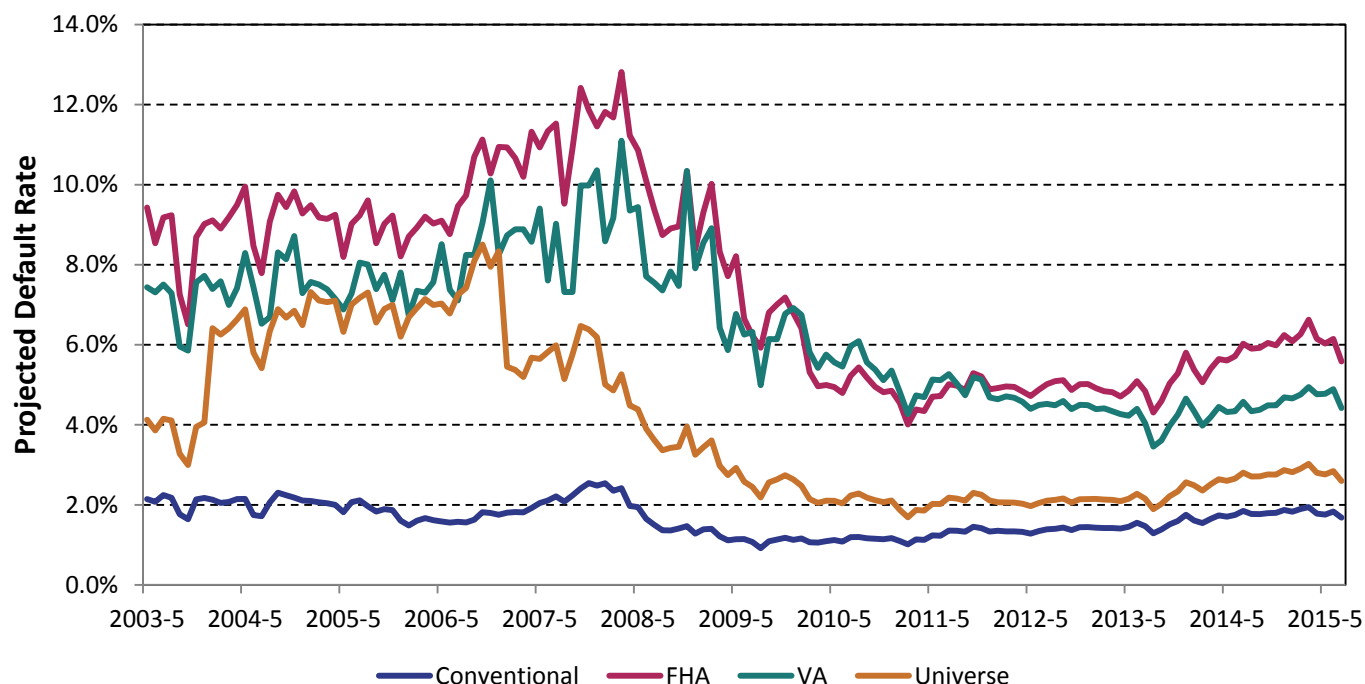


Cello's MCAI shows that, on an aggregate basis, mortgage credit availability at its 2007 peak was almost 4x its prevailing levels.

Most of the post-2007 decline in MCAI has been driven by a substantial decrease in Product Risk with a significant slump in the volume of originated Alt-A and Subprime loans.

More recently (starting 2014), there has been a gradual improvement in mortgage credit availability, but overall levels are considerably lower than in the past.

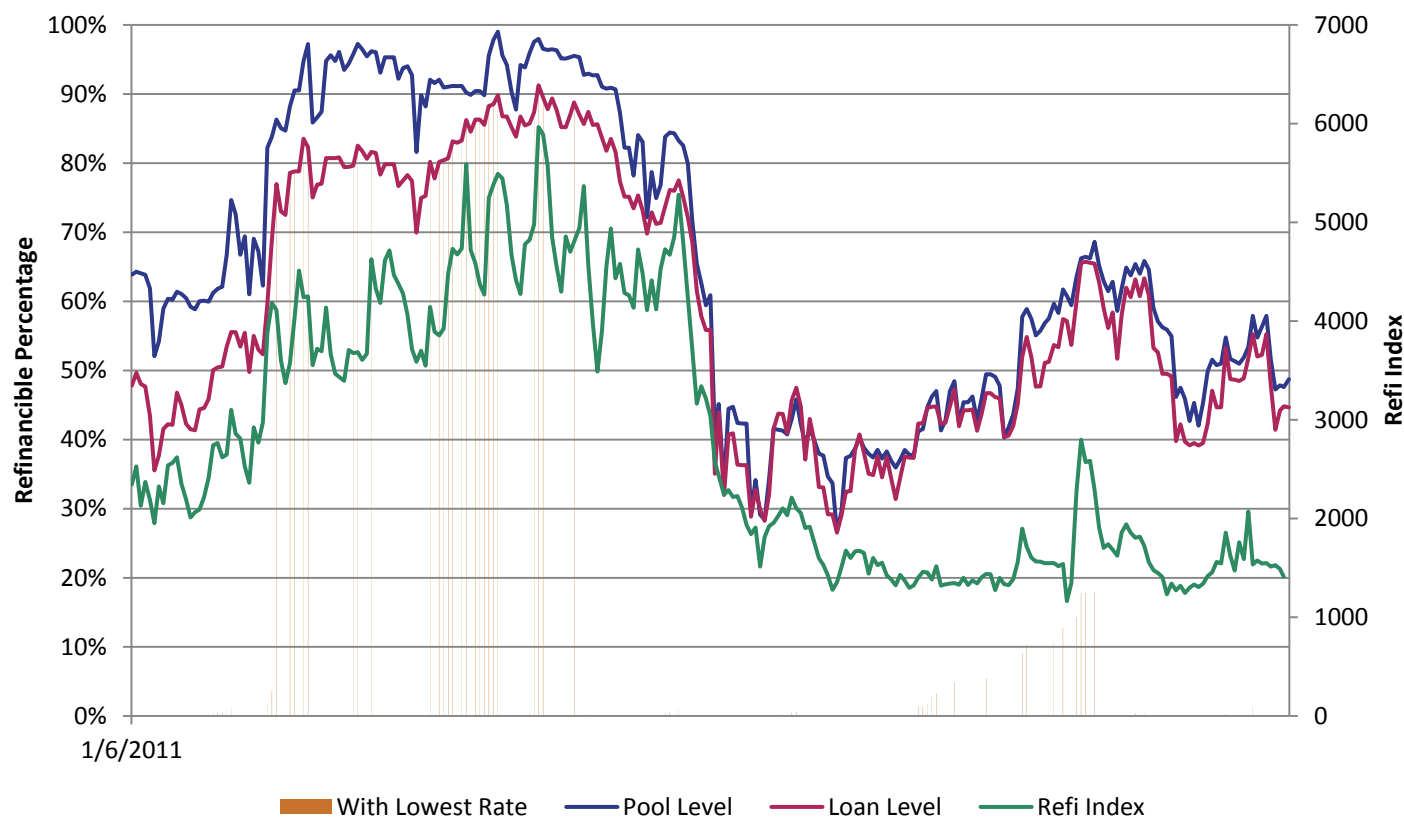
CELLO'S MORTGAGE CREDIT AVAILABILITY INDEX



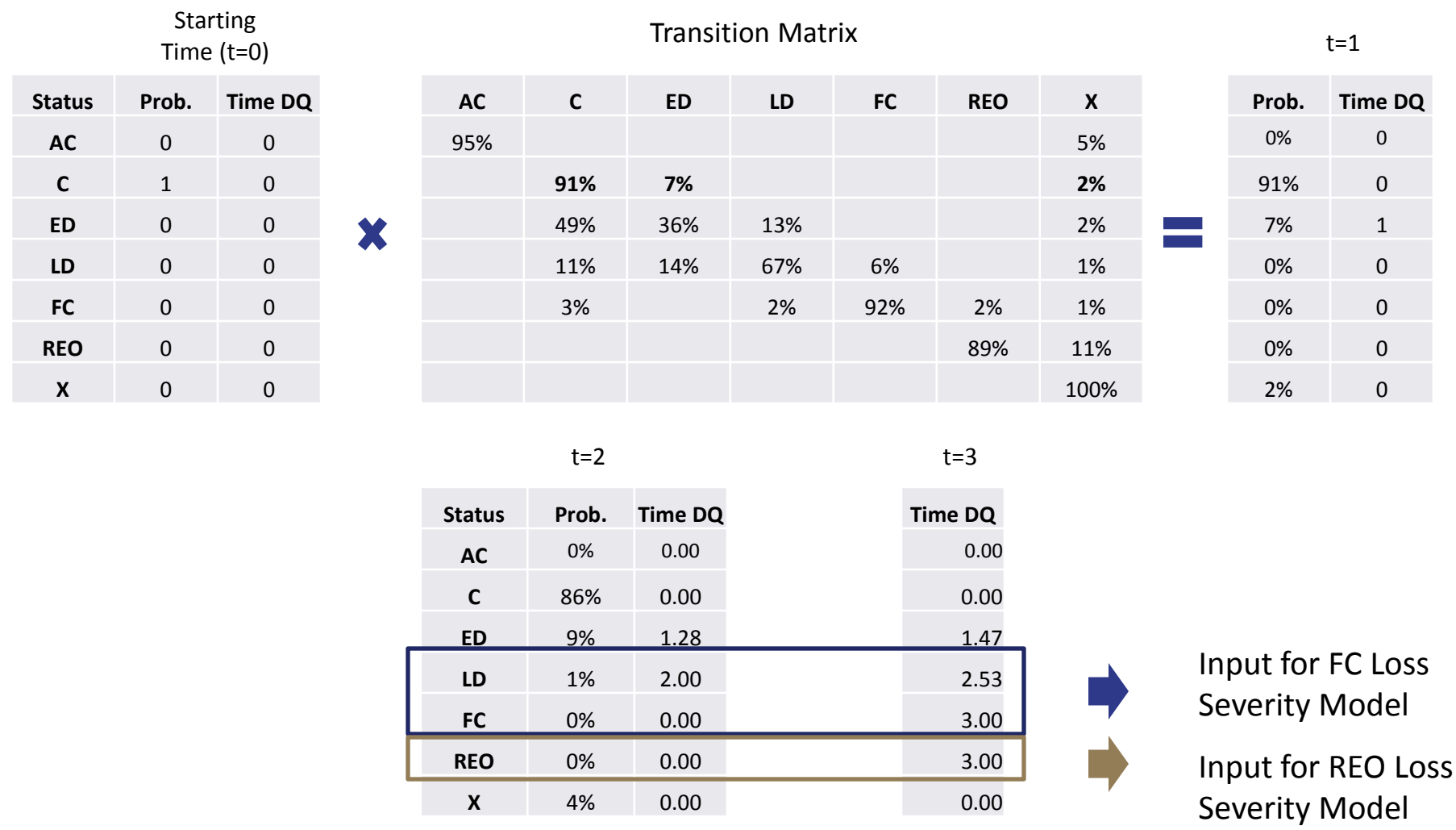
The number of borrowers who are eligible to refinance at prevailing mortgage rates and underwriting standards is a critical determinant of the level of prepayment risk in the market. Once a critical threshold of borrowers is eligible, there is often a surge in prepayment rates due to media coverage of refinancing opportunities and solicitation of refinances by originators.

Cello calculates the Refinanceable Universe in a number of different ways. In particular, the loan-level Refinanceable Universe calculation explicitly takes into account loan-level pricing adjustments and PMI-related costs.

THE REFINANCEABLE UNIVERSE VS THE REFINANCE INDEX (January 2011 – December 2015)



Stylized Single Loan Calculation



- **Scale** is built on top of **Quantlib**, a well maintained open-source C++ library for modeling, trading, and risk management. The approach allows for reducing low-level model building and focusing on complex problems.
- **Scale** allows for micro-level specification of proprietary models.
- **Scale** provides a platform for rapidly assessing new structures and strategies.
- **Scale** serves as a tangible repository for Cello IP.
- **Scale** is a training ground for new Associates/Analysts in the Research group.

THE ROLE OF TECHNOLOGY

Cello has built a comprehensive suite of trading and operations reports based on its robust databases.

Cello continues to make significant investments to build out its proprietary analytic infrastructure. The main highlights includes the following

Symphony Application

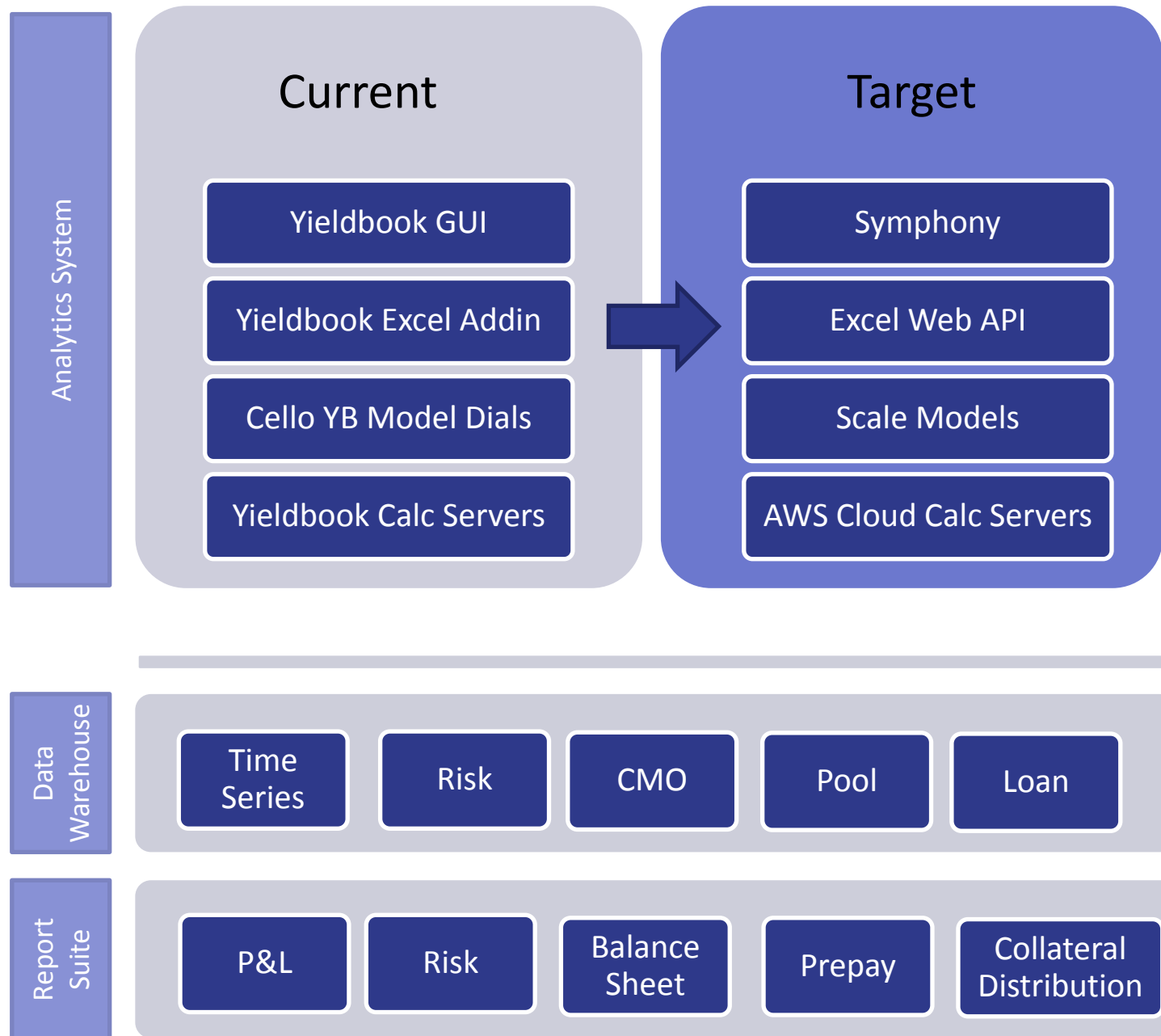
- Bond Pricing
- Collateral Analysis(slice and dice)
- Excel API

Scale Models

- Interest Rate Model
- Mortgage Rate Model
- Prepayment Model
- OAS Analytics

Motif Calculation Service

- Distributed Computing
- Amazon Cloud Integration
- In-Memory Caching
- OAS Path Distribution



TEAM BIOS

SHARAD CHAUDHARY, PHD

Head of Research

As Head of Research, Dr. Chaudhary leads the strategy and quantitative modeling effort to support investment management. Dr. Chaudhary joined Cello in 2012 with more than fifteen years of experience in fixed-income strategy, quantitative modeling and technology. Prior to 2015, when he turned his focus entirely to building out Cello's Research team, Dr. Chaudhary oversaw the continued development of Cello's operational infrastructure as Chief Operating Officer. Prior to joining Cello, Dr. Chaudhary held a range of roles at Nomura from 2009 to 2012, including running the Fixed Income Quantitative Modeling Group for the Americas, Global Head of Fixed Income Research Technology, and supporting Nomura's buildout of its Analytics infrastructure for the Securitized Products Group as a consultant. Prior to Nomura, Dr. Chaudhary was Head of Residential Securitized Products Research and Modeling at Bank of America Securities from 2000 to 2008. At Bank of America, Dr. Chaudhary built and managed a fifteen person RMBS Strategy & Modeling team from the ground-up that was ranked in the Top 5 in all RMBS categories in Institutional Investor polls by end-2008. He also headed mortgage model development and implementation for the entire spectrum of residential mortgage products.

Dr. Chaudhary earned his Ph.D. in Mathematics from Duke University and a BS degree in Mathematics and Computer Science from Lafayette College.

ROBERT MORRIS

Vice President

Mr. Morris joined Cello in May 2015. He brings experience in structured finance and in particular Non-Agency modeling. Robert worked at Citadel in the structured products group before joining Cello.

LIEN CHEN

Associate

Mr. Chen joined Cello in January 2015, with the concentration on Agency Mortgage Backed Securities strategy and quantitative modeling. Prior to joining Cello, Mr. Chen worked at Deloitte Consulting, providing financial management services with a focus on investment planning and profitability model design.

Mr. Chen earned his M.A. degree in Mathematical Finance from Columbia University, M.S. degree in Computer Science from Chinese Academy of Sciences, and B.E. degree in Electronic Engineering from Chongqing University.